

Slides 4.10: Reading and Drawing a Floor Plan to Scale, and the Design Vocabulary Check

Slide outline for Lesson 4.10. Day 1 has thirteen slides; Day 2 has ten. Keep the two conversions on the board through Lesson 4.14: 12 feet is 12 squares, and 36 inches is 3 feet is 3 squares. *[Sal: the lesson file names this deck "Slides 04.10 - Reading and Drawing a Floor Plan to Scale." The file is saved under the longer name that matches the lesson's own title, so the site links line up. Fix one or the other, not both.]*

Day 1

Slide 1: Do now

- Draw this room the way a bird would see it, looking straight down
- Put in the door, the windows, and the teacher's desk
- Two minutes. Do not worry about size yet.
- Then hold it up Notes: Sketch 1 of the stack of eleven. This is the pre-assessment: look for who draws from above and who draws a bed you can see the pillow on.

Slide 2: Two good drawings

- Both of these are good drawings
- Only one of them can prove a bed fits
- Architects, designers, firefighters, and the people who moved your family's couch all use the one drawn from above Image: two student-style drawings side by side under a document camera, one from above and one from a person's eye height. Notes: Then hold up the real floor plan. "This is somebody's actual apartment. By the end of today you will read it, and you will draw this room." Hand vote: who has used a tape measure? Who has read a map's scale bar in math or social studies?

Slide 3: Reading and Drawing a Floor Plan

- Lesson 4.10, Day 1 of 2
- Today: eleven symbols, the scale rule, and this room measured and drawn
- Tomorrow: your own bedroom, the daily sketch routine, and the Design Vocabulary Check
- Pencil only today. No pens. Image: a clean floor plan of a small apartment with a legend down the side.

Slide 4: Walls, doors, windows

- Wall: two close parallel lines, filled in
- Door: a gap in the wall, a straight line for the panel, a quarter-circle arc for the swing
- Window: a break in the wall with two thin parallel lines across it

- The swing is not decoration. It says which way the door opens and what it hits. Image: the three symbols drawn large and labeled. Notes: Draw each one on the board as you say it. Students draw it once in the box on Part 1 of the handout.

Slide 5: Closet and stairs

- Closet: a small rectangle open to the room, with a line for its door or a dashed line for a curtain
- Stairs: a run of parallel lines with an arrow and the word UP Image: both symbols drawn large and labeled.

Slide 6: The bathroom symbols

- Sink: a rectangle with a small circle for the drain
- Toilet: an oval on a small rectangle
- Tub: a long rectangle with one rounded end and a small circle
- These three are how you know a room is a bathroom with no label on it Image: the three symbols drawn large, then the same three inside a small bathroom plan.

Slide 7: The kitchen symbols, and the outlet

- Stove: a square with four small circles
- Refrigerator: a square with REF written in it
- Outlet: a short line on the wall with two dots
- Eleven symbols in all. They are the same eleven on every plan you will ever read. Image: the three symbols drawn large, then a small kitchen plan using them.

Slide 8: The scale rule

- 1/4 inch on this paper equals 1 foot in the room
- Every square on your graph paper is a quarter inch, so one square is one foot
- That is the whole rule
- 12 feet is 12 squares. 36 inches is 3 feet is 3 squares. Image: a corner of quarter-inch graph paper with a 12-square wall drawn and labeled "12 feet." Notes: Write both conversions on the board and leave them there for both days and for the rest of the unit.

Slide 9: Read a real plan

- Find the front door. Which way does it swing?
- Find a window
- Find the bathroom. How do you know?
- Find the kitchen. How do you know?
- Find a closet Image: the large real floor plan with the address removed. Notes: Students circle each item on the half-size copy. Then trace with a finger the path from the front door to the kitchen sink. Ask what the path passes through and what is in the way.

Slide 10: Zones and traffic flow

- Four zones: sleep, work, store, move
- Traffic flow: the paths people actually walk through a room
- 36 inches for any path a walker or a wheelchair uses, and for a front entry
- 30 inches everywhere else
- Both numbers stay on the board for the rest of the unit Notes: Label the four zones on the projected plan while students label their copies.

Slide 11: Measure this room

- Teams of four. One wall each, plus the door and the windows.
- Hook the tape at the corner. Read where the wall ends.
- Round to the nearest half foot: 9 feet 4 inches rounds to 9 feet 6 inches, so write 9.5
- Also measure: the door width, each window width, and the corner-to-door distance
- Every number goes in the record table. Seven minutes, then tapes down. Image: a tape measure hooked on a wall corner with the reading circled. Notes: Model the hook and the rounding once at the front before anyone stands up. Assign the walls by name. Circulate with your sticky note of the real numbers; a team more than a foot off is sent back to the wall, not corrected.

Slide 12: Now draw it

- Agree the four wall numbers as a class. They go on the board.
- Draw four walls to scale: one square is one foot
- Add the door with its swing and the windows in the right places
- Write the scale in the corner Notes: Clipboard check, two lines per team: measured to the nearest half foot and wrote it down; the drawing matches the numbers. A team that drew a 24-foot room as 24 inches gets the conversion line on the board pointed at, not read to them.

Slide 13: Hold them up, then sketch 2

- Which plan could you hand to a stranger and have them find the door?
- Sketch 2, on the back of your classroom plan: redraw this room with the desks arranged a different way
- Shade one path from the door to the window
- Two minutes Notes: Nothing is collected. The classroom plan, sketch 1, and sketch 2 go in the new project folder in order, labeled, stacked by period. That folder is turned in with the project in Lesson 4.15.

Day 2

Slide 14: Do now

- A bedroom is 10 feet by 11 feet. Draw it at 1/4 inch equals 1 foot.
- Put a 3-foot door on a short wall, with its swing

- Put a 4-foot window on a long wall
- Label the scale in the corner
- Two minutes to draw, one to trade and check: is the swing drawn, and is 10 feet 10 squares? Notes: Sketch 3 of the stack. Today's hook is one line: yesterday you measured a room somebody else built; today you draw the room you sleep in, and then you find the one thing in it that is in the way.

Slide 15: Your own bedroom, or this one

- Your choice, said out loud before anyone starts
- Draw your own bedroom from measurements you remember or estimate
- Or draw this one: 11 feet by 12 feet, a 3-foot door on a short wall, a 4-foot window centered on the far long wall, a 2 by 5 closet
- Nobody is asked to say anything about their home. No bedroom is shown to the class unless you offer it. Image: the described bedroom drawn to scale with the closet and window in place.

Slide 16: Estimating honestly

- A step is about two and a half feet
- Five steps across is about twelve feet
- An estimate you can explain beats a number you made up
- Write the four wall lengths in feet along the walls Notes: Eleven minutes on this step. Clipboard check, two lines: scale written in the corner, door swing drawn into the room.

Slide 17: Furniture at scale

- Every cut-out is drawn at the same 1/4 inch equals 1 foot, with its real size printed on it
- The cut-out and your room are the same scale, or nothing you learn here is true
- Place a bed, one storage piece, and one more piece the room needs
- Pencil the four zones: sleep, work, store, move Image: the cut-out sheet with a bed, a dresser, and a desk cut out and placed on a bedroom plan.

Slide 18: Find the pinch and fix it

- Shade every path
- Count the squares at the narrowest point. Write the width in inches.
- Find one path under 30 inches and move a piece to fix it
- One sentence at the bottom: what moved and why
- "I moved the ___ because the path was only ___ inches wide" Notes: Seven minutes. Circulate and ask the same three questions: where is your path narrowest, how many squares, how many inches is that?

Slide 19: Questions worth asking

- Your plan has a path 22 inches wide. Who in a family would that stop, and who would not even notice?
 - Why does a firefighter want a floor plan of this building, and what would they look at first?
 - A landlord hands your family a plan of an apartment. What three things would you check before signing anything?
 - Which is worse for a room: furniture that is too big, or furniture in the wrong place?
- Notes: Take one or two. These are the questions that come back on the Unit 4 Test scenario.

Slide 20: The daily floor plan do now

- Starting now and running to the last build day, every period opens with a five-minute floor plan sketch
- I put the prompt on the board. Two minutes to draw, one to check with a neighbor.
- The sheet goes in your project folder. There are eleven of them.
- You staple the stack in order, name on the first sheet, and turn it in with the project
- Each one is scored complete, partial, or missing Image: the prompt list from the handout, with sketches 1 to 4 checked off. Notes: Hand out the prompt list so an absent student can catch up. Students mark the four already done: 1 and 2 yesterday, 3 today's do now, 4 at the closure.

Slide 21: What the software does, and does not

- Four minutes on my screen: drag one wall, drop one bed
- The software does in ten seconds what you just did by hand
- It does not know the client
- Ask me what it costs and who uses it Image: a screenshot of a browser floor plan tool with one room drawn. *[Sal: name the tool your school allows; nobody but you needs a device for this.]* Notes: This is the first thing to cut if Day 2 runs long. The check is fixed at 12 minutes.

Slide 22: Sketch 4, then clear the desks

- Draw any room in this building from above, to scale, with one path shaded
- Two minutes
- One line under it: the one thing a floor plan can prove that a photograph cannot
- Then desks apart, folder closed, handouts away Notes: File sketch 4 in the folder. Then set up for the check with no talking; two minutes of transition is built into the period.

Slide 23: Design Vocabulary Check, 12 minutes

- Twelve items, twelve minutes, 24 points
- Covers Lessons 4.6 to 4.10: elements, principles, color, texture and pattern, symbols and scale
- Pencil. A calculator is allowed on items 10 and 11.



- The word bank stays on the wall. Handouts and folders stay away.
- Not sure? Write what you think and why. Notes: Hand the check out face down and read the directions aloud before anyone turns it over. Collect at the bell, or at minute 12 on the timer if the class needs the last minute to breathe. Nobody is held out of the project for this score; it tells you who needs two minutes at the elbow on day 106.